

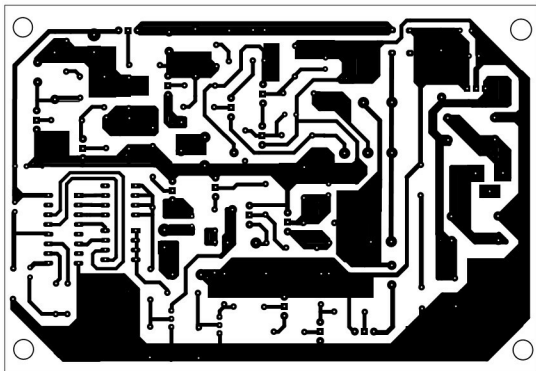


Fig. 3: An actual-size, single-side PCB for the automatic water level controller for sump/submersible pump

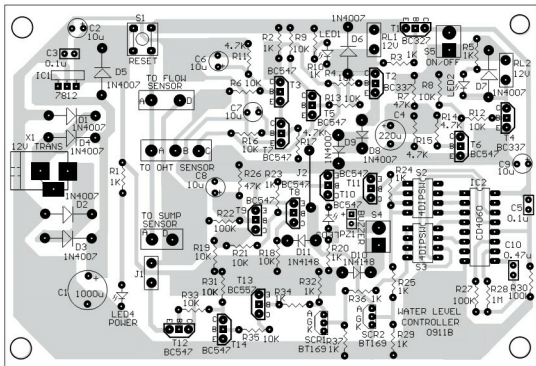


Fig. 4: Component layout for the PCB

tor T9, making it to stop conduction. Transistor T8 goes into saturation to cut off transistor T2 to stop the pump, protecting it from dry run.

Auto timer cut-off circuit. The auto timer cut-off circuit makes use of long duration outputs of 30, 60 and 120 minutes available from timer CD4060

at its pins 1, 2 and 3, respectively. The desired time is selected by DIP switch S2 to turn on the SCR2. Its anode is connected to the base of transistor T2 through resistor R4. When SCR2 is turned on, it grounds the base of transistor T2 to switch off the pump. This auto timer cut-off comes into action only when the water level sensor circuit is not switching off the pump at maximum water level due to the breakage of sensor cable or any other reason. The auto cut-off circuit also activates an alarm to indicate that the level-sensor circuit is not functioning properly. The pump will not start again till reset switch S1 is pressed. Thus the pump motor is protected from continuous run. Switch S4 is connected in series with the buzzer, which enables/disables the alarm as desired.

Submersible pump starter activation circuit. This is a simple RC time delay circuit. Resistor R7 and capacitor C3 provide a time delay (two to four second) to switch off relay RL2 connected at the collector of